

Hargurjeet Singh Ganger

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[LinkedIn](#) | [Github](#) | [Medium](#) | [Portfolio](#)

Summary

Experienced IT professional with **15+ years** in the industry, specializing in **data science, statistical analysis, and machine learning and Generative AI**. Expert in LLMs, AI model development. Proficient in **Python, SQL**, and cloud platforms like **AWS** and **GCP**, with expertise in building and deploying scalable ML and Agentic solutions.

Skills

Statistics & Machine Learning– Regression analysis, statistical inference, hypothesis testing, supervised and unsupervised learning, ensemble methods, feature engineering, boosting trees, model evaluations.

Model Evaluation & Interpretability: SHAP, LIME, Explainable AI, Precision-Recall, ROC-AUC, RMSE, MSE,, Prompt Alignment, Answer Relevancy, Faithfulness, Context Precision/Recall, Hallucination, Bias, Toxicity.

Generative AI - Vector DB, RAG patterns, Guardrails, PII Filtering, Agents, Fine-tuning, Observability, MCP

LLM Framework and API's - Langchain, Langgraph, CrewAI - tool augmented workflow, multi agent systems.

MLOps - Model deployments, Monitoring and version control, CICD pipelines, Feature store, Data drift, Model Governance, Docker, FastAPI, Gitlab, MLflow, YAML, Alerting and Logging, OpenAI, Anthropic, Gemini API's.

Cloud Platforms - AWS (SageMaker, Lambda, Bedrock, Textract, OpenSearch, Step Function, Cloud Watch)

NLP - Dense/Sparse Embeddings, Named Entity Recognition(NER), Topic Modelling, BERT/DistilBERT.

Programming & Tools: Proficient in **Python** (pandas, NumPy, Scikit-learn, TensorFlow, PyTorch), **SQL**, **PySpark**, and **SparkSQL**, **Linux** for end-to-end data workflows and machine learning pipelines.

Contributor @[stackoverflow](#), @[jovian](#) and @[PyTorch](#) communities.

WORK EXPERIENCE

Senior Data Scientist - [British Telecom](#) (BT)

(May 2022 - Present)

- Developed and deployed an AI-powered conversational chatbot using LLMs and Retrieval-Augmented Generation (RAG), enabling natural language queries and reducing manual data extraction time by 70%.
 - Built a data pipeline leveraging **AWS Textract** and **pdfplumber**, extracting contractual data with **90%+ accuracy** from PDFs, images, and scanned documents.
 - Designed scalable data pipelines with **AWS Textract**, **pdfplumber**, and **OpenSearch** to process **100K+** multimodal documents with 90%+ accuracy, integrating AWS Bedrock for seamless LLM interactions.
 - Design and implemented multistep agentic workflow using **Crewai**, integrating tool augmented pipelines for batch processing and applying **guardrails** to minimize hallucinations..
 - Developed an **LLM evaluation framework** to assess **prompt alignment, answer relevance, toxicity, bias, and hallucination detection**, leveraging the [Ragas](#) framework for structured evaluation and scoring.
 - Deployed solutions on AWS with containerisation (Docker), integrated CI/CD pipeline and implemented monitoring using custom logging to track model performance and latency tracking.
- BT** aimed to increase revenue from premium products like SD-WAN in a competitive networking market. Develop a model to identify and recommend premium products to existing customers. Engineered a **multi-label recommendation model** using **Random Forest** and **XGBoost**, analyzing customer purchase patterns and behaviors. Increased premium product sales by **10%**, directly contributing to BT's strategic objective of growing revenue and strengthening market presence.
- BT** sought to enhance customer retention and revenue by upselling Value-Added Services (VAS) to its existing customer base. Design a recommendation system to identify and promote relevant VAS offerings. Implemented a **sophisticated recommendation system** using **market basket analysis** and **apriori algorithms** to uncover actionable customer purchasing patterns. Achieved a **30% increase in VAS sales**, driving significant revenue growth and boosting BT's market share.

Data Scientist - [Royal Dutch Shell](#)

(Sep 2016 – May 2022)

- Strong experience in **Python** for data wrangling/analysis using **Pandas, Numpy, matplotlib, seaborn...**etc and built **Power BI** dashboard to forecast Materials on-time delivery in **5 geographies** within **1 year saving 10%** of the budget allocations.
- Developed **ML models (XGBoost, Random forests, Decision trees)** for predictive maintenance of types of equipment present at oil refineries resulting in 30% reduction in maintenance cost and a 25% reduction in unplanned downtime impacting workforce efficiency.
- 5+ years of Experience with databases and big data analytics technologies in the industry including **Data Warehousing/ETL** and **relational Databases**.
- Applied statistics techniques such as **classification, clustering, statistical inference**, and understanding of central tendency. Worked with DL libraries scikit-learn, TensorFlow, Keras, PyTorch...etc.
- Experience in working Agile, Scrums, Sprints...etc exposure to version control via GitHub.

IT Analyst - [TCS](#)

(Dec 2010 – Aug 2016)

- Performed **System Integration Testing & User Acceptance Testing (UAT)** to validate client PoS system.
- Spent one year in the UK guiding offshore teams with the implementation of new PoS software.
- Experience working with card and payment systems, PCI standards and ISO 8583 protocols.

Research Thesis ([View](#))

Investigated the integration of contextual language models (**GPT-3.5, Mixtral, Llama 3.1**) with classical ML models (XGBoost, Random Forest) to enhance predictive accuracy. Developed a methodology to convert tabular data into enriched textual representations and applied PCA for input optimization. Highlighting potential of LLMs in addressing data imbalance.

OPEN SOURCE DATA SCIENCE PROJECTS ([View all](#))

Production-Grade RAG Evaluation & CI Pipeline([Live app](#))

- Built a hybrid RAG system (BM25 + vector search + reranker) with automated evaluation using RAGAS (faithfulness, relevancy) and local/remote LLMs with citation grounding to reduce hallucinations.
- Implemented CI/CD quality gates via GitHub Actions to auto-run evaluations on PRs and merges when model performance drops below thresholds. The eval is run to measure faithfulness and relevance.
- Added **end-to-end observability (tracing, token usage, p50/p90 latency)** for debugging, cost control, and performance optimization using langfuse.

Local AI assistant using Ollama ([Link to the study](#))

- Built a local AI assistant using Ollama (Phi-3 Mini, Mistral 7B, LLaMA 3.2) with a fast API wrapper for on-device inference. Benchmarked model performance to analyze local inference trade-offs.
- Engineered production-grade reliability with structured JSON outputs, **Pydantic validation**, retry logic, and graceful failure handling. Conducted model comparison to evaluate consistency and performance.

EDUCATION

[LIVERPOOL JOHN MOORES UNIVERSITY](#)

2023 – 2025

M.S. in Machine Learning & Artificial Intelligence

[International Institute of Information Technology Bangalore](#)

2022 – 2023

Executive Post Graduation in Data Science and Artificial Intelligence

Courses: Statistics & Probability, Python, Machine Learning, NLP, Neural Networks, MLOps

[New Horizon College Of Engineering](#), Bangalore, India

2006 – 2010

Bachelor of Engineering, **Electronics and Communication** from **Visvesvaraya Technological University**.
